LATE BALIRAM KASHYAP GOVERNMENT MEDICAL COLLEGE

DIMRAPAL, JAGDALPUR

ACADEMIC YEAR
2026 – 27

DEPARTMENT OF PHYSIOLOGY

PROJECT ON

LIFE-SAVING MODULES

SUBMITTED BY:

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MBBS 1st Year

Batch: 2026

SUBMITTED TO:

Department of Physiology,

LT. B.R.K.M. GMC, Jagdalpur

CERTIFICATE

This is to certify that **Harsh Kumar Saw**, student of MBBS Batch 2026, LT. B.R.K.M. Government Medical College, Jagdalpur, has successfully completed the project entitled **"LIFE-SAVING MODULES"** under the Department of Physiology, during the academic year 2026 – 2027.

This project has been completed under the guidance and supervision of the Department of Physiology as a part of the academic curriculum.

Date: _____

Place: Jagdalpur

Signature of Faculty Guide

Name: _____

Designation: _____

Department of Physiology

Signature of Head of Department

Name: _____

Department of Physiology

LT. B.R.K.M, GMC, Jagdalpur

ACKNOWLEDGEMENT

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I would also like to extend my sincere thanks to my respected **Professor** and **all the faculty members of the Department of Physiology** for their valuable guidance, encouragement and assistance, which helped me in completing this project successfully.

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Cardiopulmonary Resuscitation (CPR)

1. Definition

⇒ Cardiopulmonary resuscitation (CPR) is an emergency procedure involving chest compressions, with rescue breaths often applied to maintain blood circulation and oxygen delivery to vital organs during cardiac arrest.

2. Objectives of CPR

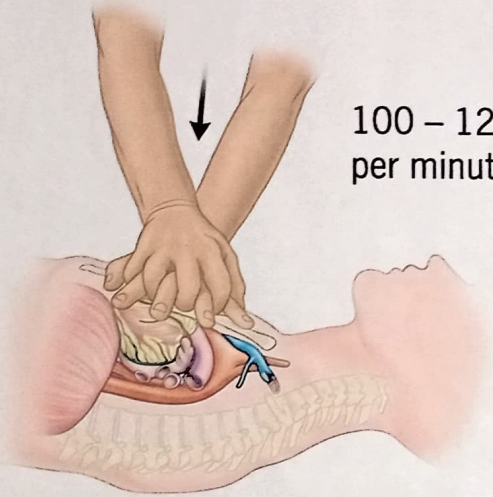
- To maintain blood flow to the brain and other vital organs
- To provide temporary circulation until advance medical care.
- To increase the chances of survival following cardiac arrest.
- To support early defibrillation when indicated.

3. Indications

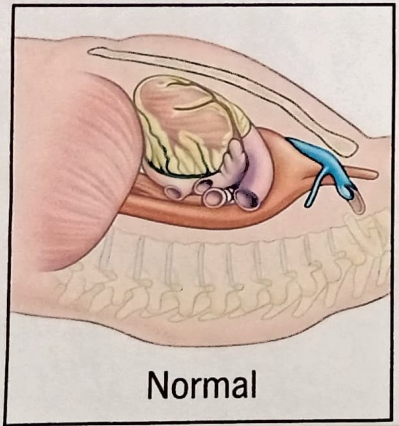
⇒ CPR is indicated when a person is unresponsive and is not breathing normally, including when only occasional gasping is present.

A lay rescuer should not delay CPR to check the pulse.

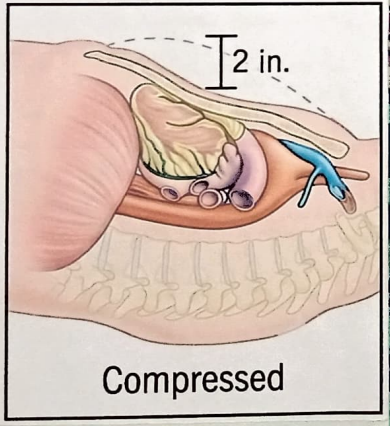
Cardiopulmonary resuscitation (CPR)



100 – 120 compressions per minute



Normal



Compressed

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4. Stages of CPR

1. Ensure safety: Check that the surrounding area is safe.

2. Check responsiveness: Tap the shoulder and ask loudly whether the person is all right

3. Call for help: Activate emergency medical services and ask someone to bring an automated external defibrillator (AED)

4. Assess breathing: Check for normal breathing for no more than 10 seconds

5. Position the patient: Place the person on their back on a firm, flat surface

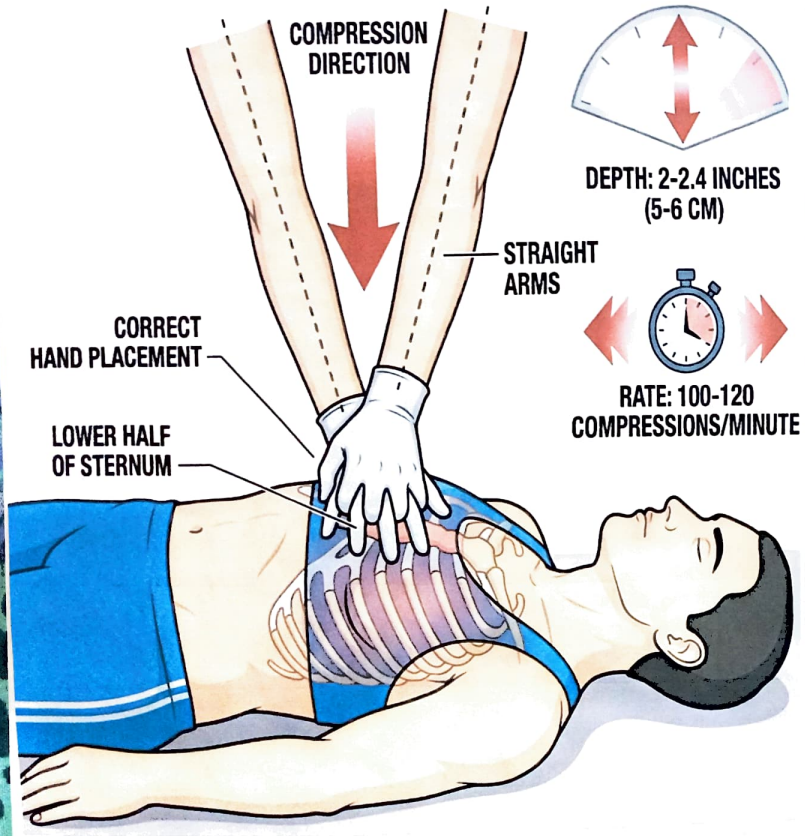
6. Perform chest compressions: Place the heel of one hand on the centre of the chest, on the lower half of the sternum, and the other hand over it.

7. Give rescue breaths: If trained and willing, provide 30 compressions followed by 2 breaths

8. Use AED: Apply it as soon as available and follow its instructions.

9. Continue resuscitation: Continue until the person shows

STANDARD CPR TECHNIQUE GUIDE



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Signs of life and breathes normally, trained help takes over, the scene becomes unsafe, or you cannot continue.

5. Characteristics of high-quality CPR

- Compression rate : 100-120 per minute
- Compression depth : 5-6 cm
- Compression-to-ventilation ratio: 30:2 for a single rescuer.
- Allow complete chest recoil after each compression.
- Minimize interruptions in chest compressions

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Heimlich manoeuvre

1. Definition

⇒ The Heimlich manoeuvre, also known as abdominal thrusts, is a first-aid technique used to relieve severe airway obstruction caused by a foreign body in a conscious adult or child older than one year.

2. Causes of choking

⇒ Choking occurs when a foreign body partially or completely obstructs the airway. Common causes include:

- Food particles such as meat or nuts.
- Small objects accidentally entering the airway
- Poorly chewed food
- Eating too quickly.

3. Signs of severe choking

- Inability to speak or breathe.
- Ineffective or absent coughing
- Clutching the throat
- Development of bluish discoloration in severe cases



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- Loss of consciousness if obstruction persists

4. Stages of the Heimlich manoeuvre

1. Recognise choking: Ask the person if they are choking.

2. Encourage coughing: If the person can cough forcefully, encourage coughing and observe closely.

3. Call for help: If coughing is ineffective or the person cannot speak or breathe, activate emergency services.

4. Give back blows: Lean the person forward and deliver 5 firm back blows between the shoulder blades.

5. Give abdominal thrusts: If the obstruction persists, stand behind the person, place a fist just above the navel and below the breastbone, grasp it with the other hand, and give 5 quick inward-and-upward thrusts.

6. Repeat: Alternate 5 back blows and 5 abdominal thrusts until the obstruction is relieved or the person becomes unresponsive.

7. If unconscious: Lower the person safely, call emergency services and begin CPR. Remove an object from the mouth only if it is clearly visible.

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Important precautions

- Do not use abdominal thrusts on infants under one year; use infant-specific back blows and chest thrusts.
- For pregnant people or when the abdomen cannot be encircled, use chest thrusts according to current first-aid guidance.
- Medical assessment may be needed after abdominal thrusts, particularly if pain or other symptoms persists.

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